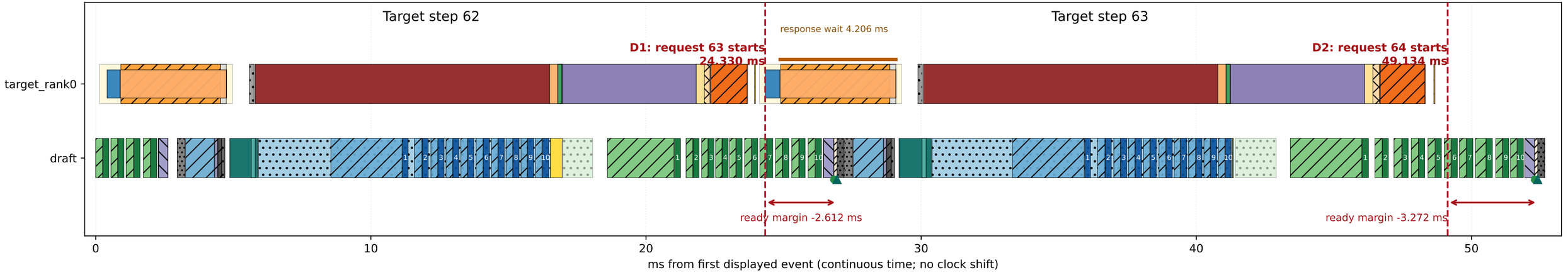
LayerSkip Llama2-7B + llama-68m | Chain; DS/PS roots40/40; K1/K2 10/10; exit21 | P1 hit -> P1 hit | UID 101_t0 | steps 62-63

Fresh local RTX4090; B1 BF16 T0.7;12 inputs x2; first10/terminal/cap/preparation excluded; separately timed native forwards | Chain; DS/PS roots40/40; K1/K2 10/10; exit21 | B=1, K1/K2 10/10

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	750	50.033	2.669	108.118
P2 hit	680	45.364	2.240	92.373
Miss	69	4.603	1.739	59.136
Overall	**1499**	**95.397**	**2.432**	**98.424**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	750	24.689	0.214	3.912	+1.773	0.214	-0.057	-2.198
P2 hit	680	24.246	0.214	3.304	+1.163	0.214	0.106	-2.037
Miss	69	29.409	0.215	3.558	+1.411	0.216	-0.140	-2.296
Overall	**1499**	**24.706**	**0.214**	**3.620**	**+1.479**	**0.214**	**0.013**	**-2.130**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = native model + LM head graph replay (output stores included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p2_cache_merge
phase1_build
phase1_prep
phase1_replay
phase2_build
phase2_prep
phase2_replay
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. Numbered blue/green bars are separately measured chain forwards.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

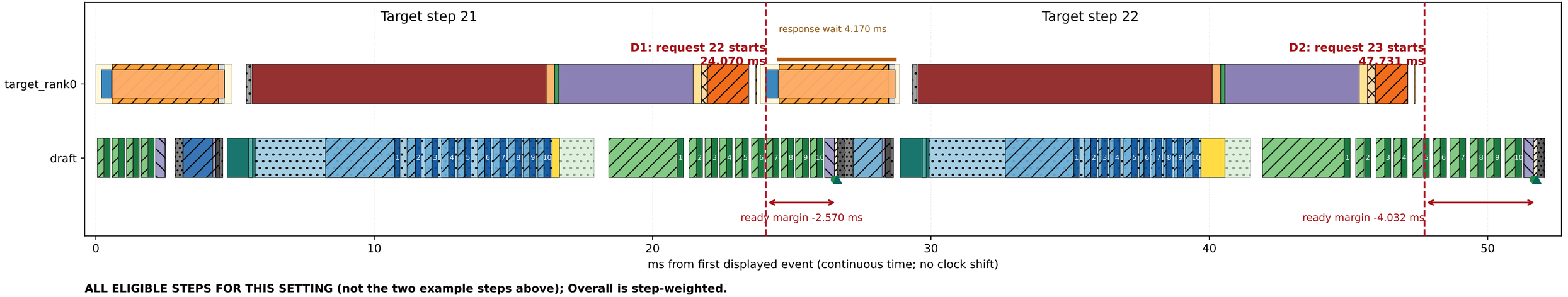
LayerSkip Llama2-7B + llama-68m | Chain; DS/PS roots40/40; K1/K2 10/10; exit21 | P2 hit -> P1 hit | UID 341_t0 | steps 21-22

Fresh local RTX4090; B1 BF16 T0.7;12 inputs x2; first10/terminal/cap/preparation excluded; separately timed native forwards | Chain; DS/PS roots40/40; K1/K2 10/10; exit21 | B=1, K1/K2 10/10

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	750	50.033	2.669	108.118
P2 hit	680	45.364	2.240	92.373
Miss	69	4.603	1.739	59.136
Overall	**1499**	**95.397**	**2.432**	**98.424**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	750	24.689	0.214	3.912	+1.773	0.214	-0.057	-2.198
P2 hit	680	24.246	0.214	3.304	+1.163	0.214	0.106	-2.037
Miss	69	29.409	0.215	3.558	+1.411	0.216	-0.140	-2.296
Overall	**1499**	**24.706**	**0.214**	**3.620**	**+1.479**	**0.214**	**0.013**	**-2.130**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = native model + LM head graph replay (output stores included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p2_cache_merge
phase1_build
phase1_prep
phase1_replay
phase2_build
phase2_prep
phase2_replay
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. Numbered blue/green bars are separately measured chain forwards.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

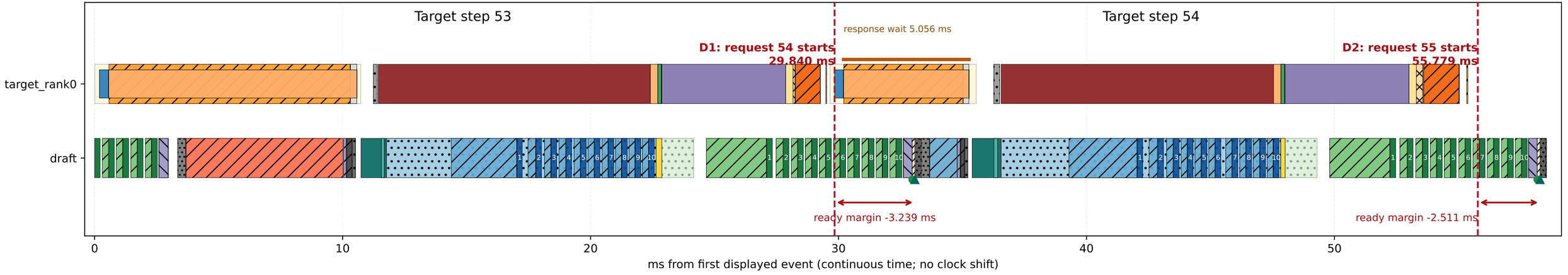
LayerSkip Llama2-7B + llama-68m | Chain; DS/PS roots40/40; K1/K2 10/10; exit21 | Miss -> P1 hit | UID 501_t0 | steps 53-54

Fresh local RTX4090; B1 BF16 T0.7;12 inputs x2; first10/terminal/cap/preparation excluded; separately timed native forwards | Chain; DS/PS roots40/40; K1/K2 10/10; exit21 | B=1, K1/K2 10/10

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

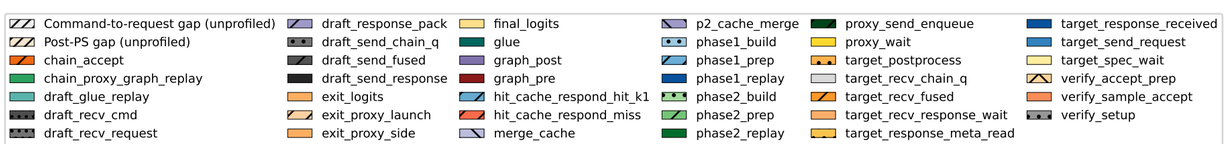
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	750	50.033	2.669	108.118
P2 hit	680	45.364	2.240	92.373
Miss	69	4.603	1.739	59.136
Overall	**1499**	**95.397**	**2.432**	**98.424**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	750	24.689	0.214	3.912	+1.773	0.214	-0.057	-2.198
P2 hit	680	24.246	0.214	3.304	+1.163	0.214	0.106	-2.037
Miss	69	29.409	0.215	3.558	+1.411	0.216	-0.140	-2.296
Overall	**1499**	**24.706**	**0.214**	**3.620**	**+1.479**	**0.214**	**0.013**	**-2.130**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = native model + LM head graph replay (output stores included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. Numbered blue/green bars are separately measured chain forwards.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

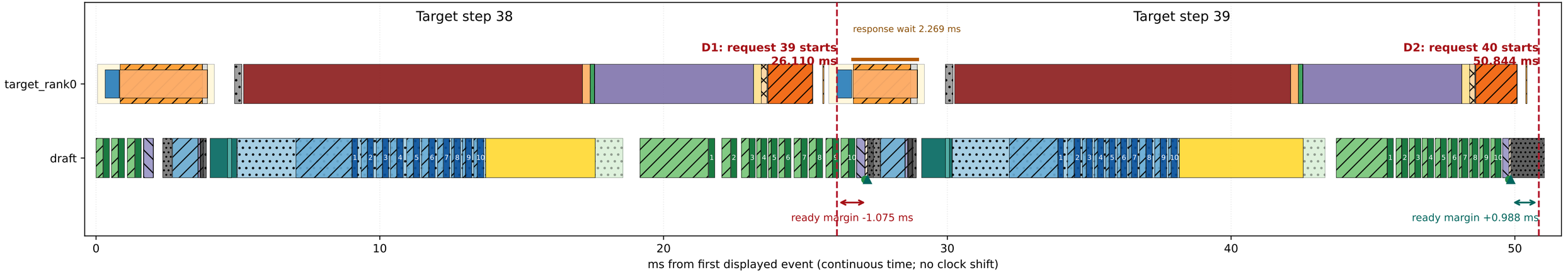
LayerSkip Llama2-13B + llama-68m | Chain; DS/PS roots40/40; K1/K2 10/10; exit26 | P1 hit -> P1 hit | UID 381_t0 | steps 38-39

Fresh local RTX4090; B1 BF16 T0.7;12 inputs x2; first10/terminal/cap/preparation excluded; separately timed native forwards | Chain; DS/PS roots40/40; K1/K2 10/10; exit26 | B=1, K1/K2 10/10

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	728	47.582	2.613	100.011
P2 hit	703	45.948	2.191	84.116
Miss	99	6.471	1.798	58.244
Overall	**1530**	**93.529**	**2.366**	**89.644**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	728	26.124	0.214	4.868	+2.733	0.213	0.769	-1.364
P2 hit	703	26.043	0.214	5.015	+2.875	0.214	1.304	-0.836
Miss	99	30.870	0.215	5.204	+3.052	0.216	1.133	-1.022
Overall	**1530**	**26.394**	**0.214**	**4.957**	**+2.819**	**0.214**	**1.039**	**-1.099**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = native model + LM head graph replay (output stores included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p2_cache_merge
phase1_build
phase1_prep
phase1_replay
phase2_build
phase2_prep
phase2_replay
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. Numbered blue/green bars are separately measured chain forwards.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

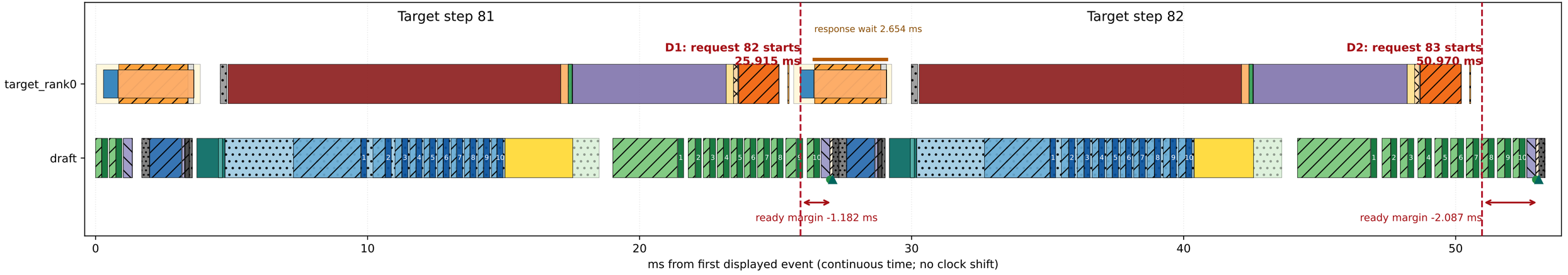
LayerSkip Llama2-13B + llama-68m | Chain; DS/PS roots40/40; K1/K2 10/10; exit26 | P2 hit -> P2 hit | UID 181_t0 | steps 81-82

Fresh local RTX4090; B1 BF16 T0.7;12 inputs x2; first10/terminal/cap/preparation excluded; separately timed native forwards | Chain; DS/PS roots40/40; K1/K2 10/10; exit26 | B=1, K1/K2 10/10

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	728	47.582	2.613	100.011
P2 hit	703	45.948	2.191	84.116
Miss	99	6.471	1.798	58.244
Overall	**1530**	**93.529**	**2.366**	**89.644**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	728	26.124	0.214	4.868	+2.733	0.213	0.769	-1.364
P2 hit	703	26.043	0.214	5.015	+2.875	0.214	1.304	-0.836
Miss	99	30.870	0.215	5.204	+3.052	0.216	1.133	-1.022
Overall	**1530**	**26.394**	**0.214**	**4.957**	**+2.819**	**0.214**	**1.039**	**-1.099**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = native model + LM head graph replay (output stores included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p2_cache_merge
phase1_build
phase1_prep
phase1_replay
phase2_build
phase2_prep
phase2_replay
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. Numbered blue/green bars are separately measured chain forwards.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

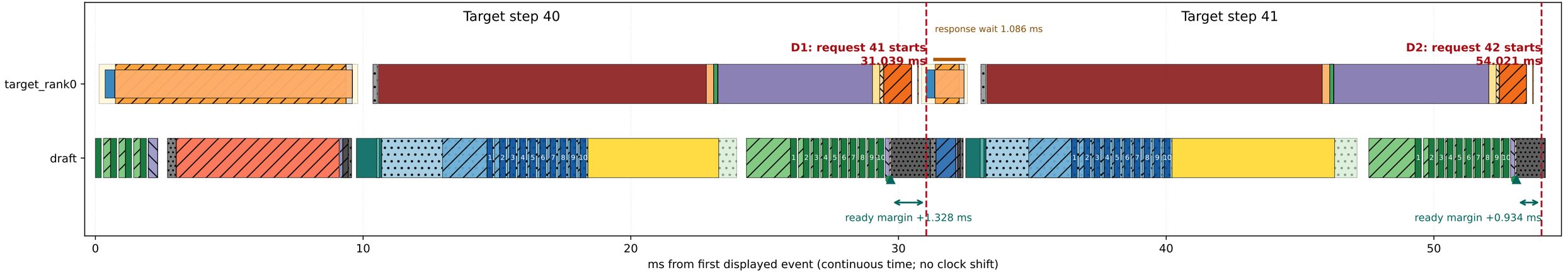
LayerSkip Llama2-13B + llama-68m | Chain; DS/PS roots40/40; K1/K2 10/10; exit26 | Miss -> P2 hit | UID 301_t0 | steps 40-41

Fresh local RTX4090; B1 BF16 T0.7;12 inputs x2; first10/terminal/cap/preparation excluded; separately timed native forwards | Chain; DS/PS roots40/40; K1/K2 10/10; exit26 | B=1, K1/K2 10/10

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

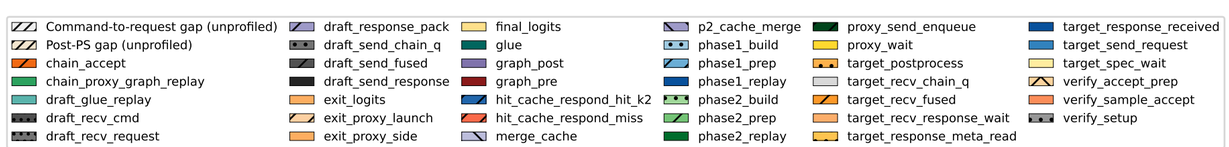
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	728	47.582	2.613	100.011
P2 hit	703	45.948	2.191	84.116
Miss	99	6.471	1.798	58.244
Overall	**1530**	**93.529**	**2.366**	**89.644**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	728	26.124	0.214	4.868	+2.733	0.213	0.769	-1.364
P2 hit	703	26.043	0.214	5.015	+2.875	0.214	1.304	-0.836
Miss	99	30.870	0.215	5.204	+3.052	0.216	1.133	-1.022
Overall	**1530**	**26.394**	**0.214**	**4.957**	**+2.819**	**0.214**	**1.039**	**-1.099**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = native model + LM head graph replay (output stores included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. Numbered blue/green bars are separately measured chain forwards.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.